

		new length = $10.0 - 2.0 = 8.0$ cm
5	B	Applying cosine rule, $(\Delta v)^2 = (20)^2 + (15)^2 - 2(20)(15)\cos 45^\circ$ $\Delta v = 14.2 \text{ m s}^{-1}$ $\Delta p = m\Delta v$ $= 0.140(14.2)$ $= 2.0 \text{ N s (2 s.f.)}$
6	C	Considering X, Y and Z as a system